r/Python • 7 mo. ago
Asleep-Organization7

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Pandas or Polars to work with dataframes?

Discussion

I've been working with Pandas long time ago and recently I noticed that Pandas 2.0.0 was released (https://pandas.pydata.org/docs/dev/whatsnew/v2.0.0.html)
However, I see lots of people pointing up that the (almost new) library Polars is much faster than Pandas.
I also did 2 analyses on this and it looks like Polars is faster:
1- https://levelup.gitconnected.com/pandas-vs-polars-vs-pandas-2-0-fight-7398055372fb
2- https://medium.com/gitconnected/pandas-vs-polars-vs-pandas-2-0-round-2-e1b9acc0f52f

What is your opinion on this? Do you like more Polars?
Do you think Pandas 2.0 will decrease the time difference between Pandas and Polars?

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draeath • 7 mo. ago

I rewrote some old code that used Pandas to graph statistics recorded at high frequency over long periods. Think around 50 columns, and millions of rows. But the catch? Each row was a span of time, not an instantaneous measurement - and the time spans were variable and overlapping. (specifically, job logs on an HPC cluster).

The pandas 1.x code took about half an hour to run - mostly because of the majority of the work being stuck in a single thread.

I rewrote it in Polars (with the help of an expert on the Polars discord) and it brought that down to *three minutes*. In contrast to the original code, this was able to leverage the cores available properly. I did not touch anything directly to do threading, like reading files in a pool or such. 100% the doing of Polars, there.

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ElasticFluffyMagnet • 7 mo. ago

That's very interesting lol.. I'm working with pandas but I think I've reached the max ceiling of optimizing it. 100+ columns and prob about a million rows. I'm gonna look into Polars now!

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nemom • 7 mo. ago

From the little I've seen, Polars looks good, but I'm sticking with Pandas for now... I do a lot of work with GeoPandas. When they release a usable version of GeoPolars, I'll take a look.

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wdroz • 7 mo. ago

> Do you think Pandas 2.0 will decrease the time difference between Pandas and Polars?

Yes, there are a lot of potential to "catch up". Pandas 2.0 and Polars both use Apache Arrow, so hopefully pandas 2.x will reduce the gap.

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NostraDavid • 7 mo. ago

And Polars also works in Rust, which Pandas doesn't, so there's a shared API there.

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saint_geser • 7 mo. ago

Personally, I switched to Polars for most tasks and rewrote a lot of codebase in Polars instead of Pandas where performance mattered. I enjoy the speed.

As for other things, Polars is relatively fresh and doesn't come with as much baggage as Pandas does so the API and workflow is more consistent.

Some things are easier done in Pandas/Numpy compared to Polars and also Pandas community is larger.

I don't think Polars is yet accepted by major machine learning libraries so if doing a lot of ML you might want to wait a bit, although you can always convert Polars DF into Pandas or Numpy.

Overall, I think Polars is better ATM.

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babygrenade • 7 mo. ago

Depends what you're doing.

If you're doing data manipulation polars will let you parallelize your operations across multiple cores.

If you're using libraries designed to work with pandas, then you'll need pandas.

If you're doing both it might make sense to do manipulations in polars and convert to pandas to use other libraries.

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lbranco93 • 7 mo. ago

This is not entirely correct. You can still do most of the work on Polars and then use `.to_pandas()` when needed

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analytics_nba • 7 mo. ago • Edited 7 mo. ago

Depends, does speed matter to you? If your workflow runs in 10 seconds, learning a new library is probably a waste of time. Do you have memory or real performance problems with pandas? Switching might be worth it (there are other alternatives as well though).

Your environment matters as well, if your company wants to use pandas for example. Personally, polars isn’t mature enough yet for me and the maintenance model isn’t really clear yet. Also, pandas is fast enough for my use case

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zaphod_pebblebrox • 7 mo. ago

At the end, for me, either of them works as long as I get paid and I can go outdoors to do my things.

In my personal experience, working with day to day data, I have not felt any difference in terms if performance. For me, tge major factor is collaboration and business requirements.

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